

Lab 05 – Prelab

Name: Abbas Huzaifa Dar, Syed Muhammad Kabir	ID: ad11013, sk11236	Section: T4
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Exercise 1:

For a loop to be synthesizable, its loop bounds must be constant, so the synthesizer knows how much replicated hardware to generate. If n is a variable or runtime signal, the iteration count is unknown at compile time, and physical hardware size cannot change dynamically. n would only be valid if it is declared as a c using parameter or localparam).

Exercise 2:

Arbitrary time delays such as #10 are for simulation only and cannot be translated into FPGA hardware. In real hardware, clock timing is supplied by dedicated FPGA clock or external clocks rather than delay statements. We take the `clk` as an input mapped to the hardware's built in clock signal.

```
always @(posedge clk) begin
```

```
    q <= d;
```

```
end
```

OR

```
always @(posedge clk_in) begin
```

```
    if (rst) clk_out <= 1'b0;
```

```
    else clk_out <= ~clk_out;
```

```
end
```

Exercise 3:

A positive-edge-triggered D Flip-Flop with a synchronous active-high enable. It is a sequential circuit and only acts on the rising edge of the clock. On the rising clock edge, the input d is captured and stored into the flip-flop output q when en is high. On the rising clock edge, the flip-flop retains its previous value when en is low.